

IIoT Network Analysis Summary Report

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1. Conceptual Overview

After reading the paper by Farag et al. (2023), it is clear that Age of Information (AoI) is different from the regular "delay" we usually talk about in networking. AoI is specifically about how fresh the data is when it gets to the controller. This is super important for what the paper calls "AoI-oriented" traffic—like a sensor monitoring a machine—where the controller needs a constant stream of new info to make sure everything is running safely. Then you have "deadline-oriented" traffic, which is more about emergency alerts. Those packets do not happen often, but when they do, they have a strict deadline to arrive or they are basically useless.

2. Data Insights

While exploring the dataset and making the graphs, I found three main things:

Crowded Networks: The heat map showed a correlation of **0.54** between the number of nodes and packet loss. It is pretty obvious that as more devices join the network, they start interfering with each other, which makes the whole thing less reliable.

Congestion Spikes: My box plots showed that even though the average AoI stays low, there are some massive spikes (outliers) that happen as you add more nodes. This tells me that the network works fine most of the time, but it can get randomly clogged up, making the data very "stale" for a moment.

The Signal Factor: Signal quality and capture thresholds are huge. If the signal is not clear enough for the receiver to "capture" it, the data gets lost, and the AoI immediately jumps up because the controller is waiting on a fresh update.

3. Machine Learning Results

I built a Random Forest model to see if we could predict the AoI based on the network settings.

Model Performance: The model actually got a negative R-squared of **-0.55**. Usually, this would mean the model is bad, but here it shows just how messy and unpredictable the simulation data is. Those huge spikes in the Age of Information are hard for a basic regression model to "guess" accurately.

What mattered most: Even with the low accuracy, the model showed that the **Capture Threshold (44%)** and the **Number of Nodes (31%)** were the biggest factors affecting the results. This matches what the research paper says about interference being the main problem.

Deep Learning Bonus: I also tried a Neural Network to predict AoI and PLP at the same time. It is a more advanced way to do it, and it showed that you could technically monitor both freshness and reliability with one model, even though it also struggled with the noisy data.

4. Strategies and Real-World Use

To help balance the trade-off between freshness and reliability, I would suggest two things:

- 1. Priority Queuing:** We should make sure "deadline" traffic (the alarms) gets to jump to the front of the line so it is not stuck behind routine status updates.
- 2. Smart Access:** Nodes should probably back off and send less often if they detect the network is getting too crowded, which would help prevent those big spikes in packet loss. These ideas are really useful for things like **Warehouse Robotics**, where robots need fresh location data so they do not hit each other, or **Factory Safety Systems**, where an emergency stop signal needs to be prioritized over everything else to keep workers safe.

5. References

Farag, H., Ali, S. M., & Stefanović, Č. (2023). On the Analysis of AoI-Reliability Tradeoff in Heterogeneous IIoT Networks. arXiv preprint arXiv:2311.13336.